

YUNSHUANG LI

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Education

University of Southern California, Ph.D. in Computer Science 08/2024 – present

Department of Computer Science, Advisor: Daniel Seita & Gaurav Sukhatme

University of Pennsylvania, M.S. in Robotics 08/2022 – 05/2024

Department of Computer and Information Science, Advisor: Dinesh Jayaraman

Zhejiang University, B.S. in Automatic Control 08/2018 – 07/2022

Chu Kochen Honors College, Dual degree in Mechanical Engineering

Publication

- 1 **BiREPO: Learning Goal-Conditioned Bimanual Nonprehensile Pose Reconfiguration Primitives on Diverse Objects**
Yunshuang Li, Xinyi Yu, Jason Chen, Bruce Zhang, Faiz ALadin, Gaurav Sukhatme, Daniel Seita
In Submission [PDF] [Website]
- 2 **ARMFUL: Learning Whole-Arm Grasp Configurations for Single-Arm Pickup of Large Objects**
Xinyi Yu, Yunshuang Li, Daniel Seita, Gaurav Sukhatme
In Submission [PDF] [Website]
- 3 **HANDFUL: Sequential Grasp-Conditioned Dexterous Manipulation with Resource Awareness**
Ethan Foong, Yunshuang Li*, Gaurav Sukhatme, Daniel Seita*
International Conference on Intelligent Robots and Systems (IROS), 2026 [PDF] [Website]
- 4 **Learning Geometry-Aware Nonprehensile Pushing and Pulling with Dexterous Hands**
Yunshuang Li, Yiyang Ling, Gaurav Sukhatme, Daniel Seita
International Conference on Robotics and Automation (ICRA), 2026 [PDF] [Website]
- 5 **Motion Capture with Millimeter-Wave Tags**
Xin Yang, Freddy Yifei Liu, Yunshuang Li, Yao Gong, Dinesh Jayaraman, Omid Abari, Mingmin Zhao
International Conference on Embedded Artificial Intelligence and Sensing Systems (SenSys), 2025 [PDF]
- 6 **Task-Oriented Hierarchical Object Decomposition for Visuomotor Control**
Jianing Qian, Yunshuang Li, Bernadette Bucher, Dinesh Jayaraman
8th Annual Conference on Robot Learning (CoRL), 2024 [PDF] [Website]
- 7 **DROID: A Large-Scale In-the-Wild Robot Manipulation Dataset**
Alexander Khazatsky, Karl Pertsch* (et al., including Yunshuang Li)*
Robotics: Science and System (RSS), 2024 [PDF] [Website]
- 8 **Universal Visual Decomposer: Long-Horizon Manipulation Made Easy**
Zichen Zhang, Yunshuang Li*, Osbert Bastani, Abhishek Gupta, Dinesh Jayaraman, Yecheng Jason Ma†, Luca Weihs†*
CoRL Learning Effective Abstractions for Planning workshop (Oral, Best Paper Award), 2023;

NeurIPS Foundation Models for Decision Making workshop (**Oral**), 2023;
 International Conference on Robotics and Automation (ICRA), 2024 (**Best Paper Finalist in Computer Vision**) [PDF] [Website]

- 9 **Open X-Embodiment: Robotic learning datasets and RT-X models**
Large collaboration (et al., including Yunshuang Li)
 International Conference on Robotics and Automation (ICRA), 2024 (**Best Conference Paper Award**) [PDF] [Website]

- 10 **Im2Contact: Vision-Based Contact Localization Without Touch or Force Sensing**
Leon Kim, Yunshuang Li, Michael Posa, Dinesh Jayaraman
 7th Annual Conference on Robot Learning (CoRL), 2023 [PDF]

- 11 **PEg TRAnsfer Workflow recognition challenge report: Does multi-modal data improve recognition?**
Arnaud Huaulmé, Kanako Harada, (et al., including Yunshuang Li)
 Computer Methods and Programs in Biomedicine, 2023 [PDF]

- 12 **Control of Pneumatic Artificial Muscles with SNN-based Cerebellar-like Model**
Hongbo Zhang, Yunshuang Li*, Yipin Guo*, Xinyi Chen, Qingyuan Ren*
 International Conference on Social Robotics (ICSR), 2021 [PDF]

- 13 **Collaborative Recognition of Feasible Region with Aerial and Ground Robots through DPCN**
Yunshuang Li, Zheyuan Huang, Zexi Chen, Yue Wang, Rong Xiong
 International Conference on Real-time Computing and Robotics (RCAR), 2021 [PDF]

Awards

USC Viterbi School of Engineering Fellowship	2024
USC Women in Science and Engineering Graduate Top-Off Award	2024
CoRL Travel Grant	2023 & 2024
Outstanding Masters Student Researcher Award, SEAS, UPenn	2024
Best Conference Paper Award, ICRA	2024
Best Paper Finalist in Computer Vision, ICRA	2024
Best Paper Award at CoRL LEAP workshop	2023
President Gutmann Leadership Award, administered by GAPSA, UPenn	2023
Chiang Chen Oversea Graduate Scholarship	2022
Outstanding Graduate of Zhejiang Province	2022
National Scholarship issued by Ministry of Education of the People's Republic of China	2021
Champion of International Robotic Workflow Recognition Challenge in MICCAI 2021	2021

Industry Experience

Samsung Research America , Robot Intelligence Lab <i>Research Intern</i>	06/2026 – Present
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Lab Lead: Kris Hauser; Mentor: Devesh Jha

Develop multimodal world action model(WAM) on Sharpa dexterous hands for Samsung logistics tasks.

Service

Conference Reviewer: CoRL, ICRA, IROS

Journal Reviewer: T-RL, RAL

Workshop Reviewer: NeurIPS, ICRA, RSS

Teaching Assistant for CSCI 103 - Introduction to Programming, at USC. Spring 2025

Teaching Assistant for CIS 5190 - Applied Machine Learning, at UPenn. Spring 2024

Teaching Assistant for CIS 5200 - Machine Learning, at UPenn. Fall 2023

Teaching Assistant for MEAM 5200 - Introduction to Robotics, at UPenn. Spring 2023

Mentorship

Ethan Foong, Undergrad from Northwestern University Next:

Miscellaneous Experience

Volunteer for Robotic Science and System 2025 June 2025

Board Member of Penn Chinese Dance Club 2023

Volunteer for 5th Annual Learning for Dynamics & Control Conference (L4DC) June 2023

Instructor of Python Club at Carver Engineering and Science High School, Philadelphia Spring 2023

Volunteer for GRASP ROBO Master's Student Open House March 2023

Volunteer Instructor for a Rural High School, Yunnan, China Summer 2020

Technical Skills

Languages: Python, MATLAB, C, C++, HTML/CSS, L^AT_EX

Developer Tools: VS Code, Anaconda, Docker

Technologies/Frameworks: Linux, ROS, PyTorch, Tensorflow, AutoCAD, Solidworks, GitHub